

RL-ESG Portfolio Optimization

via Daily Sentiment, CVaR, and PPO

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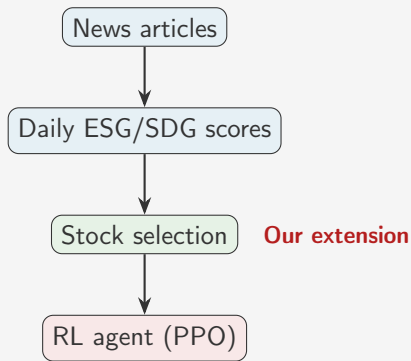
Problem Statement

Motivation

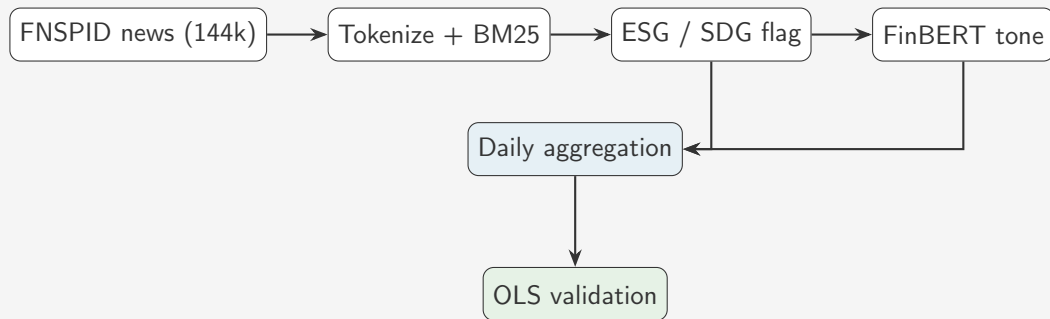
- ▶ Feng et al. (2024): daily ESG/SDG sentiment + monthly ML optimization.
- ▶ But: lexicon NLP, symmetric variance, no daily adaptation.

Our Goal

1. Upgrade sentiment with **FinBERT** + **BM25** topic tagging.
2. Replace static ML with a **daily RL agent** using CVaR reward.
3. **Diagnose** why RL fails/succeeds out-of-sample.



Sentiment Pipeline



$$r_{t+1} = \beta_0 + \beta_1 \cdot \text{Score}_t + \beta_2 \cdot \text{Mkt_RF}_t$$

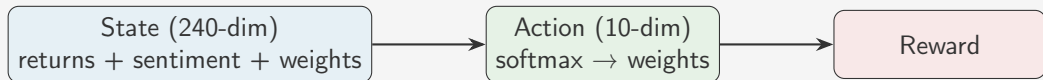
overall ($\beta_1 = 0.0014^{***}$), **SDG** ($\beta_1 = 0.0064^{***}$)

Two-Stage Stock Selection (Monthly)



$$\text{Factor score} = 0.80 \cdot \text{rank}(\alpha^{\text{FF3}}) + 0.20 \cdot \text{rank}(-|\beta - 1|)$$

RL Environment (Daily Rebalancing)

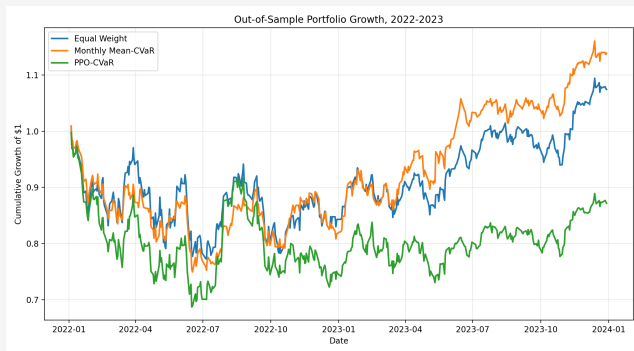


$$R_t = w^\top r_{t+1} - 2.0 \text{CVaR}_{0.95} + 0.25 (\text{upside}) - 0.001 |w - w_{\text{prev}}|$$

Training: PPO (100k steps, 2019-2021). Monthly episodes with re-selection at month start.

Out-of-Sample Performance (2022-2023)

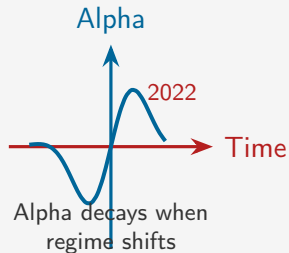
Strategy	Ann.Ret.	Vol.	Sharpe	Sortino	Turnover
Monthly Mean-CVaR	7.1%	20.6%	0.436	0.657	0.049
Equal Weight (sel.)	3.9%	25.2%	0.276	0.402	0.000
PPO-CVaR	-7.0%	25.9%	-0.150	-0.231	0.732



Why Did RL Fail?

Diagnostics

- ▶ **Selection itself is neutral:** EW on selected 10 Sharpe = 0.276, same as full universe.
- ▶ **Alpha decay:** historical α not predictive in 2022 – a regime shift.
- ▶ **Overtrading:** turnover 0.73/day $\rightarrow$ rapid reallocation amplifies losses.



Proposed Improvements

Short-term fixes

Trend filter

Volatility-scaled α

Long-term research

Turnover budget

Online learning

Regime-switch RL

Ensemble of policies

Conclusion & Takeaways

1. FinBERT + BM25 daily sentiment **works** – especially SDG news predicts returns.
2. Static selection is **not harmful**; the RL agent's weighting caused losses.
3. **Regime shifts** break historical alpha signals → RL trained in a bull market fails in a bear market.
4. **Overtrading** amplifies losses; tighter penalty or volatility scaling needed.
5. Negative result → **constructive research** – we now know *what not to do* and have clear fixes.

“A precise measurement of what doesn't work is also progress.”

Thank you!

Questions?